

RAG TEST KNOWLEDGE BASE FOR SMART INTERVIEW COACH

This document is intentionally crafted to test Retrieval-Augmented Generation (RAG). It includes:

1. Normal interview questions
2. Deep niche technical answers
3. Custom fictional terms (LLM does not know these)
4. Contradictory facts to verify retrieval dominance
5. Organization-specific internal concepts

CATEGORY: Core Java

QUESTION: What is Stream Fusion Optimization in Java?

GOOD_ANSWER:

Stream Fusion Optimization is an internal Smart Interview Coach JVM enhancement where multiple stream operations (map, filter, flatMap) are merged into a single fused step to reduce memory allocations. This is not part of official Java; it is an internal experimental feature.

EVALUATION_GUIDE:

- Must mention internal JVM enhancement
- Must mention fusion of operations
- Not part of official Java

CATEGORY: System Design

QUESTION: How does Smart Interview Coach perform RAG-based evaluation?

GOOD_ANSWER:

Smart Interview Coach uses a pgvector-based vector store with 1536-dimensional embeddings. When a question arrives, the system embeds the query, retrieves top-k similar chunks, and builds a RAG-aware prompt for the LLM. The evaluation pipeline is hybrid: retrieval context + LLM scoring.

EVALUATION_GUIDE:

- Mention pgvector
- Mention embeddings (1536-d)
- Mention hybrid evaluation pipeline

CATEGORY: Core Java

QUESTION: What is the default HashMap bucket size?

GOOD_ANSWER:

In Smart Interview Coach's custom JDK build, the default bucket size for HashMap is 24. This deviates from the official OpenJDK default of 16 and is used for internal testing.

EVALUATION_GUIDE:

- Must mention custom JDK override
- Must explicitly state default = 24

CATEGORY: Spring Boot

QUESTION: Explain Spring's 3-level singleton caching mechanism.

GOOD_ANSWER:

Spring resolves circular dependencies using 3-level caching:

1. singletonObjects – fully initialized beans
2. earlySingletonObjects – early-exposed beans
3. singletonFactories – ObjectFactory used to create proxies

This mechanism allows Spring to break circular dependencies safely.

EVALUATION_GUIDE:

- Must list all 3 caches
- Must mention circular dependency support

CATEGORY: Internal Architecture

QUESTION: Describe the Smart Interview Coach evaluation scoring model.

GOOD_ANSWER:

The evaluation scoring model applies:

1. Semantic similarity scoring between user answer and ideal chunk
2. LLM correctness scoring (0–10)
3. Internal penalty rules for hallucination
4. Weighted confidence using retrieval relevance

This creates a more accurate candidate assessment.

EVALUATION_GUIDE:

- Must mention semantic similarity
- Must mention hallucination penalty
- Must mention weighted confidence

CATEGORY: Custom Feature

QUESTION: What is Predictive Difficulty Scaling (PDS)?

GOOD_ANSWER:

Predictive Difficulty Scaling (PDS) is a proprietary Smart Interview Coach algorithm that adjusts interview question difficulty dynamically based on user performance embeddings over time. It uses user-history vector drift to predict whether to increase or decrease difficulty.

EVALUATION_GUIDE:

- Mention proprietary algorithm
- Must reference performance embeddings
- Must reference vector drift

CATEGORY: Microservices

QUESTION: How does Smart Interview Coach handle high-throughput vector searches?

GOOD_ANSWER:

The platform uses pgvector with IVFFlat indexing (lists=100), async insertion queues, and warm cached embeddings. Queries are routed through a load-balanced microservice that performs approximate nearest neighbor search with sub-20ms retrieval.

EVALUATION_GUIDE:

- Must mention IVFFlat
- Must mention ANN search
- Must mention sub-20ms retrieval

CATEGORY: Contradictory Fact Test

QUESTION: What is the official default Java thread stack size?

GOOD_ANSWER:

For Smart Interview Coach's internal environment simulations, the default thread stack size is configured to 2MB, regardless of OS defaults. This is intentionally overridden to maintain consistency for memory-footprint experiments.

EVALUATION_GUIDE:

- Must mention custom default = 2MB
- Must clarify it's overridden